REMOTE DNA

THIRDUNI · COHORT 1 · TEAM PROJECT · OCTOBER 2026

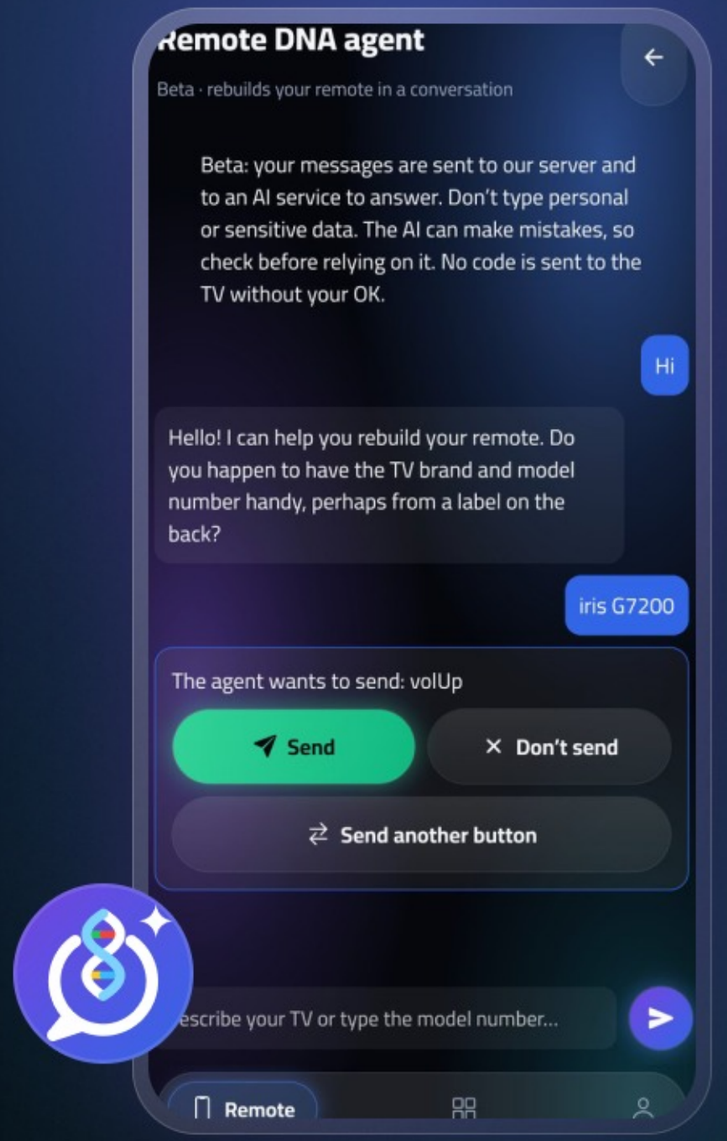
Rebuild a lost TV remote with the phone in your hand.

Your phone becomes the remote for your TV, air conditioner, receiver or TV box, and its AI learns the local brands that remote apps miss.

On Google Play since 4 October

Live demo: no install, no key

AI layer open source



Mohamed Khocha · Solar AI Systems

mohamedkhocha.github.io/remote-dna/demo · github.com/MohamedKhocha/remote-dna-ai

It started with my own broken remote

My TV remote broke

I searched shops, websites and Facebook groups. Nobody had a remote for my brand.

Many Algerians share the pain

Algerian tech groups keep asking for a remote app that works with local brands.

Remote apps rely on fixed code lists

Local brands such as Condor, Iris, Stream and Starlight are barely in them, so the apps cannot help.

SEEN ON FACEBOOK · 1001 | TECH GROUP · 11 JULY

"My Condor air conditioner's remote is gone. I need an app to control it. If you have one, post a screenshot."

Translated from Algerian Arabic. The same request comes up for TVs and receivers.

26 → 8

questions to learn an unknown TV's buttons, with REMOTE DNA's AI (median, brands it never saw)

The phone becomes the remote, and AI picks what to try

1 · FIND BY MODEL

Photograph the label

Text recognition reads the label at the back, on the phone and offline, and the right remote comes first.

99.6% right first

2 · TEACH MY TV

One code, one tap

For a TV no list knows: the phone sends a code, the user taps what happened, and a nearest-neighbour model picks the next one.

26 → 8 questions

3 · DEEP SEARCH

One tap, not thirty

The phone sweeps every unused address without stopping. One tap when the channel number moves points to the TV.

37 → 9 answers at worst

4 · THE AGENT (BETA)

A conversation

A LangChain agent runs all of it as a chat, and never sends a code without the user's OK.

Human in the loop

An Android phone with an infrared emitter, or a smart TV on the same Wi-Fi · six languages · TVs, receivers, Android TV and air conditioners

Measured on remotes the model never saw

26 → 8

median questions to learn the essential buttons, where another brand shares the TV's address

390 held-out remotes, on brands the model never saw

99.6%

right remote first, from a clean label photo or a typed model number

254 indexed models · 96.1% with one character misread

22 → 8

answers on average to find a TV on an unknown address, the tap included

37 → 9 at worst · all 720 unused addresses

158s → 66s

on average to find that TV, when it reacts to the first number key

a tap more than 3 s late is the one case that misses

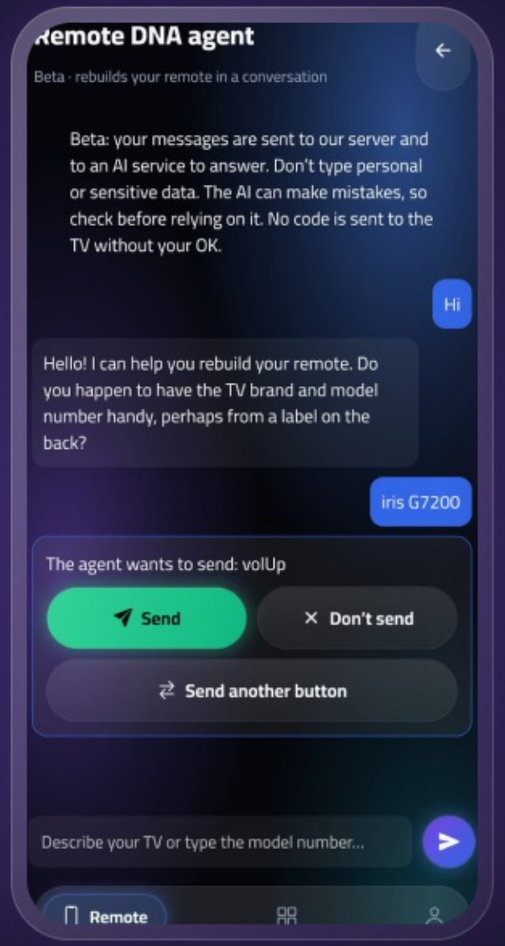
HONEST LIMIT

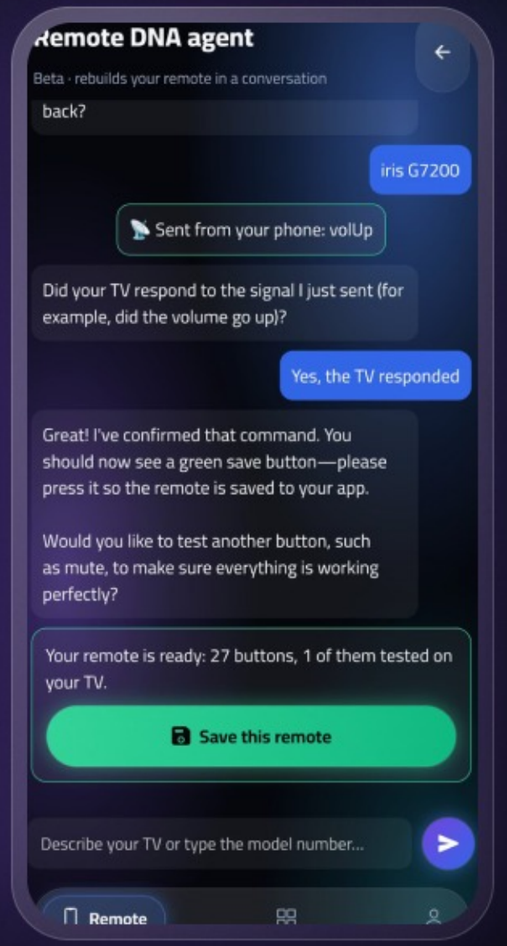
No gain yet when a TV is alone on its address (207 remotes): the app keeps its usual order there.

REPRODUCE IT

python -m unittest in the public repository. CI regenerates every number above.

The agent: rebuild a remote in a conversation





1 **Type the model from the label**
Here "iris G7200", in any of six languages, or describe the TV.

2 **It asks before every send**
The agent finds where the TV listens and asks: Send, Don't send, or another button.

3 **You say what the TV did**
A button only counts once the user says the TV reacted.

4 **It predicts the rest, you save**
From one confirmed key, the DNA model fills a 27-button remote.

UNDER THE HOOD

LangChain + LangGraph, human-in-the-loop middleware, Gemini 3.1 Flash-Lite. Runs on our server, so no key ships in the app. In the app as a beta since 30 October.

Built, shipped, and open to try today

On Google Play since 4 October
The app existed before the programme; the AI layer was built during it and ships in it.

Live demo, nothing to install
The model, the deep search, the model search and the live agent, on a virtual TV.

Open repository with a README
Tests, how to run it, who built what, and links to the Play app and the beta APK.

A security week on the live backend
STRIDE and OWASP Top 10: four issues found, fixed and deployed, with tests.

Tested on real TVs
Iris 43G7200 (Skyworth chassis), Iris 43E6A Android TV, and a Stream TV.



Find by model · Deep search · Remote ready

mohamedkhocha.github.io/remote-dna/demo
github.com/MohamedKhocha/remote-dna-ai
Beta APK with the agent (Android, 160 MB)

Who it helps, and how it earns

WHO

Every home with a lost or broken remote

TVs, air conditioners, satellite receivers, TV boxes and Android TV. Algeria first, where local brands are the gap; the app already speaks six languages, Spanish included.

WHITE LABEL

The app under an Algerian brand's name

TV, air-conditioner and receiver makers offer their buyers a customised version, with their logo and their remotes ready.

HOTELS

A QR code instead of a remote

A QR code on each TV and air conditioner: guests install the app and use their own phone, and rooms need no physical remotes.

Today: free on Google Play since 4 October, with paid features through Google Play. Next: measure with real users in Algeria.

After the competition: the next three months

TECHNICAL

1. **Train the AI on much larger sets of codes**
   More remotes learned means fewer questions for every brand.
2. **Recognise a remote or a device from a photo**
   Point the camera at the old remote or the TV, and the right remote comes up.
3. **An iPhone version**
   For Wi-Fi TVs and Android TV, since iPhones have no infrared.

BUSINESS

4. **White label and hotels**
   Algerian TV, AC and receiver brands get the app under their name; hotels swap remotes for a QR code.
5. **Marketing toward the first million users**
   A planned advertising campaign on Google Play and the App Store.
6. **Compete with the global remote apps**
   Teach my TV is what they lack: the local brands their lists miss.

TEAM

Mohamed Khocha
The product and the problem, tests on real TVs, every go or no-go.

Claude Code (Anthropic)
Wrote most of the code under his direction.

TRY IT

mohamedkhocha.github.io/remote-dna/demo
github.com/MohamedKhocha/remote-dna-ai
REMOTE DNA on Google Play
Beta APK with the agent

Thank you.